

Leveraging the Target-Decoy Competition Setup to Bound the False Discovery Rate with Dependent p -values

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1 Introduction

1.1 Classical FDR Control

Hypothesis testing is a prominent statistical inference method used to determine if there is sufficient evidence to reject a scientific hypothesis. It involves the comparison of two hypotheses, the null hypothesis and the alternative hypothesis, and the calculation of a corresponding p -value p , which is the probability that, under the null, the test statistic is as extreme as observed. Then, the null hypothesis is rejected if $p \leq \alpha$, where α is the significance level.

A natural extension of this method is multiple hypothesis testing, where m hypotheses $(H_i)_{i \in [m]}$ are tested simultaneously, with $[m] = 1, 2, \dots, m$ and i the index of hypothesis H_i . Then many of the null hypotheses will be true, but some will be false. Let us denote the indices of the true null hypotheses as $\mathcal{H}_0$ and the indices of the false null hypotheses as $\mathcal{H}_1$.

Suppose that for $i \in \mathcal{H}_0$, the p -values have a $U(0, 1)$ distribution. Then, if $i \in \mathcal{H}_0$ for every $i \in [m]$, there is a probability α that each true null is falsely rejected, leading to an expected number of αm falsely rejected true nulls.

Suppose that we have m_0 true null hypotheses of the m total hypotheses. We then introduce some notation in the table below.

Table 1: Outcomes of simultaneously testing m hypotheses

	Not significant	Significant	Total
True null hypotheses	U	V	m_0
False null hypotheses	T	S	$m - m_0$
Total	$m - R$	R	m

In particular, the random variable R is the number of significant hypotheses or *rejections*, and the random variable V is the number of true null hypotheses we reject, or the *false discoveries*.

The **False Discovery Proportion** (FDP) is defined to be

$$\text{FDP} := V / \max \{R, 1\}$$

and we define the **False Discovery Rate** (FDR) as the expectation of this random variable:

$$\text{FDR} = \mathbb{E}[\text{FDP}] = \mathbb{E}[V / \max \{R, 1\}].$$

Controlling the FDR has been a long-standing interest in the statistical community, dating back to Benjamini and Hochberg [2], who proposed a procedure that controls the FDR at level $(m_0/m)\alpha \leq \alpha$, where α is the desired FDR threshold. This procedure controls the FDR under either the independence of p -values or positive regression dependency on a subset (PRDS). Subsequently, Storey et al. [10] proposed another procedure using a q -value approach.

However, the problem becomes more complex under arbitrary local dependence of true null p -values, and these two procedures often fail to control the FDR. Benjamini and Yekutieli [3] proposed a general procedure that controls the FDR under arbitrary dependence. This procedure is much more conservative and hence far less statistically powerful, as it includes a factor of H_m , the m -th harmonic number.

1.2 Target-Decoy Competition

Target-Decoy Competition (TDC) is a standard method for providing a conservative estimate of the FDR, originating from the mass spectrometry community as a way to conduct hypothesis testing without p -values (Elias and Gygi [5]). In this procedure, for each hypothesis H_i , a *Target score* is given and one corresponding *Decoy score* is generated from the true null distribution. The larger of these two becomes the “winning score” W_i and the label $L_i = 1$ if the Target score is larger, and $L_i = -1$ if the Decoy score is larger. Multiple Decoy scores can be generated for each Target score if needed.

We call hypotheses with $L_i = 1$ *Target wins* and those with $L_i = -1$ *Decoy wins*. Define $T(x)$ and $D(x)$ as the numbers of Target and Decoy wins respectively with score $W_i \geq x$. He et al. [8] establish that, under the assumption that for $i \in \mathcal{H}_0$,

$$\mathbb{P}(L_i = -1) = r\mathbb{P}(L_i = 1)$$

where $r > 0$ is the number of Decoy scores per Target score, a conservative estimate for the FDR is

$$\widehat{\text{FDR}}(x) = \frac{D(x)}{r[\max \{T(x), 1\}]} \quad (1)$$

such that for any $x \in \mathbb{R}$,

$$\mathbb{E}[\widehat{\text{FDR}}(x)] \geq \text{FDR}(x),$$

where $\text{FDR}(x)$ is the FDR of the Target wins with $W_i \geq x$.

A variant of this estimate, also provided by He et al. [8], can control the FDR, but only under independence or suitable exchangeability conditions. Currently, there is no general analogue of Benjamini–Yekutieli’s arbitrary-dependence guarantee in the classical TDC framework. Relatedly, in regression, the *knockoff* framework of Barber and Candès [1] compares each feature to a synthetic “knockoff” copy and selects by a competition rule, achieving finite-sample FDR control without p -values.

This paper aims to develop a procedure that provides an upper bound on the FDR, rather than one that controls it. The remainder of the paper is structured as follows. In [Section 2](#), we discuss the bounding procedure and a mapping from p -values to the competition setup. In [Section 3](#), we discuss some theoretical properties of our bound, and in [Section 4](#), we analyse simulations on independent and dependent p -value structures and compare our procedure with classical FDR control procedures.

2 The Proposed Procedure

The **FDR Bound** procedure leverages the TDC procedure to provide an upper bound on the FDR rather than to control it.

For each hypothesis $(H_i)_{i \in [n]}$, we compute a target score and two decoy scores. The sets of target, first-decoy, and second-decoy scores are mutually independent, but may exhibit arbitrary dependence within each set.

Then, each hypothesis has a winning score W_{1i} and a hidden label $L_i^h \in \{1, 0, -1\}$. As in TDC, we define the sets $T := \{i \in [n] : L_i^h = 1\}$ of *Targets*, $ED := \{i \in [n] : L_i^h = -1\}$ of *Estimating Decoys*, and $BD := \{i \in [n] : L_i^h = 0\}$ of *Bounding Decoys*. The Estimating Decoys will be used to compute the FDR Bound, while the Bounding Decoys are used to determine the cutoff for the selected Targets and Decoys.

Then, each hypothesis also has a second winning score W_{2i} and an observed label $L_i^o \in \{1, -1\}$. We define $T_2 := \{i \in [n] : L_i^o = 1\}$ as the *Pseudo-Targets* and $D_2 := \{i \in [n] : L_i^o = -1\}$.

We want to use a similar FDR estimation function based on (1). To that end, for $m \in [n]$ define the *FDR estimation function*

$$\omega(m) := \frac{(\mathbf{D}(m) + c)/r}{\mathbf{T}(m) - \mathbf{D}(m)/r} \quad (2)$$

where

$$\mathbf{D}(m) := \#\{i \in [m] : L_i^o = -1\}, \quad \mathbf{T}(m) := \#\{i \in [m] : L_i^o = 1\} \quad (3)$$

are the running totals of T_2 and D_2 over the first m indices.

We define r as

$$r := \frac{\mathbb{E}[\#BD]}{\mathbb{E}[\#ED]} \quad (4)$$

which we use to control the ratio of Bounding Decoys to Estimating Decoys.

The numerator $(\mathbf{D}(m) + c)/r$ is an estimate of the number of selected Decoys $D(x)$ as defined in (1). Here, we add a constant c to emulate what is done in TDC to prevent an overly enthusiastic threshold for a small number of discoveries, which would often lead to a higher FDP and hence a higher FDR Bound. He et al. [8] also show that in the TDC setup, adding $c = 1$ to the numerator of (1) is needed for finite-sample FDR control, since the current estimator only provides asymptotic FDR control.

The denominator $\mathbf{T}(m) - \mathbf{D}(m)/r$ is an estimate of the number of selected Targets $T(x)$ as defined in (1). Note that ω increases with respect to $\mathbf{D}$ and decreases with respect to $\mathbf{T}$.

2.1 FDR Bound

The procedure takes the n tuples $(L_i^h, W_{1i}, L_i^o, W_{2i})_{i \in [n]}$ as input, and it begins by sorting the tuples in descending order of W_{2i} . We define $\mathbf{T}(m)$ and $\mathbf{D}(m)$ as in (3) and use the FDR estimation function ω defined in (2) to find the *cutoff*

$$\gamma^* := \max_{m \in [n]} \{\omega(m) \leq \alpha\},$$

where α is the desired FDR threshold. This γ^* is a standard step-up choice to maximise the number of discoveries, while keeping the FDR estimate $\leq \alpha$.

Let $A = \{i \in [\gamma^*] : L_i^o = 1\}$ be the set of indices with $L_i^o = 1$ in the top- γ^* tuples, and define

$$T_S := A \cap T, \quad D_S := A \cap ED$$

as the Target wins and Decoy wins in the top- γ^* tuples. These are the selected Targets and Decoys.

Finally, we return the FDR Bound as

$$\text{FDR}_B(\gamma^*) = \frac{\#D_S}{\max\{\#T_S, 1\}},$$

and rejections as T_S .

In simulation, we typically clip the FDR Bound at 1 (by taking $\min\{\text{FDR}_B, 1\}$) since the FDR cannot be greater than 1, so the bound would be meaningless if it did. In many cases, the actual/raw FDR Bound and the clipped value coincide; however, in some cases, the raw FDR Bound is well beyond the clipped value (see [Section 4.3](#)).

We denote this procedure as `FDR_Bound_Procedure` (Algorithm 1).

Algorithm 1: FDR_Bound_Procedure

Data: The n tuples (L^h, W_1, L^o, W_2) from Algorithm 2; desired FDR α ; r ; c

Result: The FDR Bound FDR_B and rejections T_S

- 1 Sort the tuples in descending order of W_2
 - 2 Compute cumulative sums of $\#\{L^o[i] = 1\}$ and $\#\{L^o[i] = -1\}$ and store these in arrays **T** and **D**
 - 3 Initialise the array ω of size n
 - 4 **for** $i \leftarrow 1$ **to** n **do**
 - 5 Set $\omega[i] = \frac{(\mathbf{D}[i] + c)/r}{\mathbf{T}[i] - \mathbf{D}[i]/r}$
 - 6 **end**
 - 7 Find $x = \max_{i \in [n]} \{\omega[i] \leq \alpha\}$
 - 8 Find the selected Targets and Estimating Decoys as
 $T_S = \{i \in [x] : L^o[i] = 1, L^h[i] = 1\}, \quad D_S = \{i \in [x] : L^o[i] = 1, L^h[i] = -1\}$
 - 9 Return the **FDR Bound** $\text{FDR}_B = \frac{\#D_S}{\max\{\#T_S, 1\}}$ and **selected Targets** T_S
-

2.2 p -values to Scores and Labels

The FDR Bound procedure itself does not strictly require p -values; however, in our main setup in this paper, we begin with a number of true null and false null hypotheses with corresponding p -values.

To facilitate the use of the FDR Bound procedure, we use the following transformation to process these p -values into a TDC-based setup with labels and scores. This procedure is inspired by a similar transformation initially proposed in Lei and Fithian [9], which allowed them to apply the TDC setup in the p -value context.

Let $(H_i)_{i \in [n]}$ be hypotheses, with $i \in \mathcal{H}_0$ for true nulls and $i \in \mathcal{H}_1$ for false nulls, and $\mathcal{H}_0 \cap \mathcal{H}_1 = \emptyset$. Associated with each hypothesis H_i is a corresponding p -value p_i , and for $i \in \mathcal{H}_0$, $p_i \sim \text{U}(0, 1)$ or stochastically dominates the $\text{U}(0, 1)$ distribution.

Fix $\tau \in (0, \frac{1}{2})$ and partition the interval $[0, 1]$ into $[0, \tau] \cup (\tau, 1 - \tau) \cup [1 - \tau, 1]$. The hidden label L^h is given as

$$L_i^h := \begin{cases} 1, & \text{if } p_i \in [0, \tau] \\ 0, & \text{if } p_i \in (\tau, 1 - \tau) \\ -1, & \text{if } p_i \in [1 - \tau, 1] \end{cases}.$$

We denote the *Targets* as $T := \{i \in [n] : L_i^h = 1\}$, *Bounding Decoys* as $BD := \{i \in [n] : L_i^h = 0\}$ and *Estimating Decoys* as $ED := \{i \in [n] : L_i^h = -1\}$. The observed

label L^o is then given as

$$L_i^o := \begin{cases} 1, & \text{if } L_i^h \in \{1, -1\} \\ -1, & \text{if } L_i^h = 0 \end{cases}$$

and we similarly denote the *Pseudo-Targets* as $T_2 := \{i \in [n] : L_i^o = 1\}$ and $D_2 := \{i \in [n] : L_i^o = -1\}$.

To generate suitable scores for the FDR Bound procedure from the p -values, the interval $[0, 1]$ is folded twice, and two scores are generated as described below.

- The first fold maps the p -values in $[\frac{1}{2}, 1]$ to $[0, \frac{1}{2}]$ in a mirrored manner, so that the p -values of the hypotheses with index $i \in ED$ now lie in the same interval as those with $i \in T$, namely $[0, \tau]$. The p -values of the hypotheses with $i \in BD$ now end up in $[\tau, \frac{1}{2}]$. Define the associated scores as

$$\widetilde{W}_{1i} := \min(p_i, 1 - p_i).$$

- The second fold maps the masked p -values (that is, the values after the first fold) in $[\tau, \frac{1}{2}]$ to $[0, \tau]$ using the affine map

$$s_\tau(x) = \tau \cdot \frac{1/2 - x}{1/2 - \tau}. \quad (5)$$

Thus, the p -values of the hypotheses with $i \in T \cup ED$ stay the same in this second fold, whereas those with $i \in BD$ are mapped inversely again. Define the associated scores as

$$\widetilde{W}_{2i} := \begin{cases} \widetilde{W}_{1i}, & \text{if } L_i^o = 1 \\ s_\tau(\widetilde{W}_{1i}), & \text{if } L_i^o = -1 \end{cases}.$$

However, we note that these scores must be flipped so that the larger scores are preferred in the procedure. Thus, we finally define

$$W_{1i} := 1 - \widetilde{W}_{1i}, \quad W_{2i} := 1 - \widetilde{W}_{2i}.$$

Then, the tuples $(L_i^h, W_{1i}, L_i^o, W_{2i})_{i \in [n]}$ represent the complete data for the FDR Bound procedure if we started from a p -value setting. We will refer to this procedure as `p-values_to_Scores_and_Labels` (Algorithm 2).

Algorithm 2: p-values_to_Scores_and_Labels

Data: An array of p -values $(p_i)_{i \in [n]}$ corresponding to n hypotheses $(H_i)_{i \in [n]}$;

$$\tau \in (0, \frac{1}{2})$$

Result: n tuples of scores and labels $(L^h[i], W_1[i], L^o[i], W_2[i])$ for $i \in [n]$

1 Initialise arrays L^h, W_1, L^o, W_2 each of length n

2 **for** $i \leftarrow 1$ **to** n **do**

3 Set the hidden labels

$$L^h[i] = \begin{cases} 1, & \text{if } p[i] \in [0, \tau] \\ 0, & \text{if } p[i] \in (\tau, 1 - \tau) \\ -1, & \text{if } p[i] \in [1 - \tau, 1] \end{cases}$$

4 Compute the first fold scores

$$\widetilde{W}_1[i] = \min(p[i], 1 - p[i])$$

5 Set the observed labels

$$L^o[i] = \begin{cases} 1, & \text{if } L^h[i] \in \{1, -1\} \\ -1, & \text{if } L^h[i] = 0 \end{cases}$$

6 Compute the second fold scores using (5)

$$\widetilde{W}_2[i] = \begin{cases} \widetilde{W}_1[i], & \text{if } L^o[i] = 1 \\ s_\tau(\widetilde{W}_1[i]), & \text{if } L^o[i] = -1 \end{cases}$$

7 Flip the scores to obtain the winning scores

$$W_1[i] = 1 - \widetilde{W}_1[i], \quad W_2[i] = 1 - \widetilde{W}_2[i]$$

8 **end**

9 Return the tuples $(L^h[i], W_1[i], L^o[i], W_2[i])$ for $i \in [n]$

3 Theoretical Analysis

He et al. [8] establish the conservatism of the standard TDC estimator in (1) for the FDR under the (generalised) Equal Chance Assumption. Each hypothesis in TDC yields a single winning score W_i and label $L_i \in \{1, -1\}$ and for $i \in \mathcal{H}_0$, conditional on the σ -field

$$\mathcal{F} = \sigma(\{W_i\}_i, \{L_i\}_{i \in \mathcal{H}_1}, \{i\}_{i \in \mathcal{H}_1})$$

generated by the information on scores W_i and any information on false null hypotheses, they assume that

$$\mathbb{P}(L_i = -1 \mid \mathcal{F}) = r\mathbb{P}(L_i = 1 \mid \mathcal{F}), \quad r > 0.$$

The case $r = 1$ is called the **Equal Chance Assumption** (ECA). We now aim to establish our main result using a similar assumption.

3.1 Our Main Result

We will assume throughout that the tuples have been sorted by W_2 in descending order, and that there are n hypotheses. Let

$$\mathcal{F} = \sigma(\{W_{1i}\}_i, \{W_{2i}\}_i, \{L_i^o\}_i, \{i\}_{i \in \mathcal{H}_1}, \{L_i^h\}_{i \in \mathcal{H}_1}) \quad (6)$$

be the partial information containing the observed labels and the scores (and hence ranks) of the tuples in the output of Algorithm 2, and any information on $i \in \mathcal{H}_1$, including their identities and labels. Then, we assume that the following holds.

Assumption 1. For index $i \in [n]$ with $L_i^o = 1$ and $i \in \mathcal{H}_0$,

$$\mathbb{P}(L_i^h = 1 \mid \mathcal{F}) = \mathbb{P}(L_i^h = -1 \mid \mathcal{F}).$$

In other words, if a true null hypothesis is a Pseudo-Target, then it is equally likely to be a Target or an Estimating Decoy.

Under this assumption, we will state and prove the main theorem of this paper.

Theorem 1 (FDR Bound). Denote $\text{FDR}_B(m)$ and $\text{FDP}(m)$ by the FDR Bound and FDP obtained by selecting the targets in the top- m tuples. If $\gamma^* = \max \{m : \omega(m) \leq \alpha\}$, then under Assumption 1,

$$\mathbb{E}[\text{FDP}(\gamma^*)] \leq \mathbb{E}[\text{FDR}_B(\gamma^*)].$$

In other words, our procedure provides a valid average upper bound on the FDR.

Proof. We begin by fixing $m \in [n]$ and defining the following variables of m .

Table 2: Summary of notation for counts among the top- m tuples

Variable	Definition	Interpretation
$PT(m)$	$\{i \in [m] : L_i^o = 1\}$	Pseudo-Targets in the top- m tuples
$N_0(m)$	$\#\{i \in PT(m) : i \in \mathcal{H}_0\}$	# of true null hypotheses
$T_1(m)$	$\#\{i \in PT(m) : i \in \mathcal{H}_1, L_i^h = 1\}$	# of true discoveries
$ED_1(m)$	$\#\{i \in PT(m) : i \in \mathcal{H}_1, L_i^h = -1\}$	# of false null Estimating Decoys
$T_0(m)$	$\#\{i \in PT(m) : i \in \mathcal{H}_0, L_i^h = 1\}$	# of false discoveries
$ED_0(m)$	$N_0(m) - T_0(m)$	# of true null Estimating Decoys
$T^*(m)$	$T_0(m) + T_1(m)$	# of selected Targets
$D^*(m)$	$ED_0(m) + ED_1(m)$	# of selected Decoys

Since L_i^h is not $\mathcal{F}$ -measurable for $i \in \mathcal{H}_0$, it follows that $T_0(m)$ and hence the last four variables in Table 2 are not $\mathcal{F}$ -measurable.

Expressing $\text{FDP}(m)$ and $\text{FDR}_B(m)$ in terms of these variables we have

$$\text{FDP}(m) = \frac{T_0(m)}{\max\{T^*(m), 1\}}, \quad \text{FDR}_B(m) = \frac{D^*(m)}{\max\{T^*(m), 1\}}. \quad (7)$$

We first show that Assumption 1 implies the following equation.

$$\mathbb{E}[T_0(m) \mid \mathcal{F}] = \mathbb{E}[ED_0(m) \mid \mathcal{F}] \quad (8)$$

By definition, we can write $T_0(m) = \sum_{i=1}^m \mathbf{1}_{A_i}$, where we define the events $A_i := \{i \in \mathcal{H}_0, L_i^h = 1, L_i^o = 1\}$, and $\mathbf{1}_{(\cdot)}$ is the indicator function. Hence, we have

$$\mathbb{E}[T_0(m) \mid \mathcal{F}] = \mathbb{E}\left[\sum_{i=1}^m \mathbf{1}_{A_i} \mid \mathcal{F}\right] = \sum_{i=1}^m \mathbb{E}[\mathbf{1}_{A_i} \mid \mathcal{F}]$$

using the linearity of conditional expectation.

By the fact that $\mathbb{E}[\mathbf{1}_{A_i} \mid \mathcal{F}] = \mathbb{P}(A_i \mid \mathcal{F})$, this gives us

$$\mathbb{E}[T_0(m) \mid \mathcal{F}] = \sum_{i=1}^m \mathbb{P}(A_i \mid \mathcal{F}).$$

Similarly, we write $ED_0(m) := \sum_{i=1}^m \mathbf{1}_{B_i}$ where $B_i := \{i \in \mathcal{H}_0, L_i^h = -1, L_i^o = 1\}$. Using Assumption 1 and the definitions of A_i and B_i , it follows that $\mathbb{P}(A_i \mid \mathcal{F}) = \mathbb{P}(B_i \mid \mathcal{F})$ for each $i \in [m]$. Thus, we continue the equality to obtain

$$\begin{aligned} \mathbb{E}[T_0(m) \mid \mathcal{F}] &= \sum_{i=1}^m \mathbb{P}(A_i \mid \mathcal{F}) = \sum_{i=1}^m \mathbb{P}(B_i \mid \mathcal{F}) \\ &= \sum_{i=1}^m \mathbb{E}[\mathbf{1}_{B_i} \mid \mathcal{F}] = \mathbb{E}\left[\sum_{i=1}^m \mathbf{1}_{B_i} \mid \mathcal{F}\right] \\ &= \mathbb{E}[ED_0(m) \mid \mathcal{F}]. \end{aligned}$$

Since $N_0(m)$ is $\mathcal{F}$ -measurable, then by (8),

$$\begin{aligned} N_0(m) &= \mathbb{E}[N_0(m) \mid \mathcal{F}] \\ &= \mathbb{E}[T_0(m) + ED_0(m) \mid \mathcal{F}] \\ &= 2\mathbb{E}[T_0(m) \mid \mathcal{F}]. \end{aligned} \tag{9}$$

Next, define the function

$$g(x) := \frac{N_0(m) - 2x + ED_1(m)}{\max\{x + T_1(m), 1\}},$$

and observe that using the definitions in Table 2 and (7), we have

$$\text{FDR}_B(m) - \text{FDP}(m) = \frac{D^*(m) - T_0(m)}{\max\{T^*(m), 1\}}$$

with numerator

$$\begin{aligned} D^*(m) - T_0(m) &= ED_0(m) + ED_1(m) - T_0(m) \\ &= N_0(m) - T_0(m) + ED_1(m) - T_0(m) \\ &= N_0(m) + ED_1(m) - 2T_0(m) \end{aligned}$$

and denominator

$$\max\{T^*(m), 1\} = \max\{T_0(m) + T_1(m), 1\}.$$

Hence, plugging in $x = T_0(m)$ we obtain

$$g(T_0(m)) = \frac{N_0(m) - 2T_0(m) + ED_1(m)}{\max\{T_0(m) + T_1(m), 1\}} = \text{FDR}_B(m) - \text{FDP}(m). \quad (10)$$

We define $h(x) := \frac{1}{\max\{x + T_1(m), 1\}}$ and $f(x) := N_0(m) - 2x$ and can rewrite $g(x)$ as

$$g(x) = \frac{N_0(m) - 2x}{\max\{x + T_1(m), 1\}} + \frac{ED_1(m)}{\max\{x + T_1(m), 1\}} = h(x)f(x) + ED_1(m)h(x).$$

For each $k, j \in \mathbb{N}$, we also define the deterministic functions

$$h_k(x) := \frac{1}{\max\{x + k, 1\}}, \quad f_j(x) := j - 2x.$$

Then, we note that

$$h'_k(x) = \begin{cases} 0, & \text{if } x + k \leq 1, \\ -\frac{1}{(x+k)^2}, & \text{if } x + k > 1 \end{cases}$$

and $f'_j(x) = -2$. Thus, both are monotone non-increasing functions. For the random variable $X := T_0(m)$, we have $0 \leq h_k(X) \leq 1$ and hence $\mathbb{E}|h_k(X)| < \infty$. We also have that $0 \leq X \leq m$, hence

$$|f_j(X)| = |j - 2X| \leq j + 2X \leq j + 2m,$$

so that $\mathbb{E}|f_j(X)| < \infty$ as well. Consequently, $|h_k(X)f_j(X)| \leq |f_j(X)|$ and so $\mathbb{E}|h_k(X)f_j(X)| < \infty$.

Therefore, we can apply the conditional Chebyshev's association inequality (Golikova and Kruglov [6]) on h_k and f_j ; that is,

$$\mathbb{E}[h_k(X)f_j(X) \mid \mathcal{F}] \geq \mathbb{E}[h_k(X) \mid \mathcal{F}] \cdot \mathbb{E}[f_j(X) \mid \mathcal{F}] \quad \forall k, j \in \mathbb{N}.$$

Writing $T_1 = T_1(m)$ and $N_0 = N_0(m)$ it follows that

$$(\mathbf{1}_{\{T_1=k\}} \cdot \mathbf{1}_{\{N_0=j\}}) \mathbb{E}[h_k(X)f_j(X) \mid \mathcal{F}] \geq \mathbf{1}_{\{T_1=k\}} \mathbb{E}[h_k(X) \mid \mathcal{F}] \cdot \mathbf{1}_{\{N_0=j\}} \mathbb{E}[f_j(X) \mid \mathcal{F}].$$

Since T_1 and N_0 are $\mathcal{F}$ -measurable, this is equivalent to

$$\mathbb{E}[\mathbf{1}_{\{T_1=k\}} h_k(X) \cdot \mathbf{1}_{\{N_0=j\}} f_j(X) \mid \mathcal{F}] \geq \mathbb{E}[\mathbf{1}_{\{T_1=k\}} h_k(X) \mid \mathcal{F}] \cdot \mathbb{E}[\mathbf{1}_{\{N_0=j\}} f_j(X) \mid \mathcal{F}].$$

Note that $\mathbf{1}_{\{T_1=k\}} h_k(X) = \mathbf{1}_{\{T_1=k\}} h(X)$ and $\mathbf{1}_{\{N_0=j\}} f_j(X) = \mathbf{1}_{\{N_0=j\}} f(X)$ for any

$k, j \in \mathbb{N}$ by construction. Hence, we obtain the inequality

$$\mathbb{E}[\mathbf{1}_{\{T_1=k\}}h(X) \cdot \mathbf{1}_{\{N_0=j\}}f(X) \mid \mathcal{F}] \geq \mathbb{E}[\mathbf{1}_{\{T_1=k\}}h(X) \mid \mathcal{F}] \cdot \mathbb{E}[\mathbf{1}_{\{N_0=j\}}f(X) \mid \mathcal{F}],$$

and therefore

$$\mathbf{1}_{\{T_1=k\}} \cdot \mathbf{1}_{\{N_0=j\}} \cdot \mathbb{E}[h(X)f(X) \mid \mathcal{F}] \geq \mathbf{1}_{\{T_1=k\}}\mathbb{E}[h(X) \mid \mathcal{F}] \cdot \mathbf{1}_{\{N_0=j\}}\mathbb{E}[f(X) \mid \mathcal{F}].$$

Summing over all $k, j \in \mathbb{N}$, we finally get

$$\mathbb{E}[h(X)f(X) \mid \mathcal{F}] \geq \mathbb{E}[h(X) \mid \mathcal{F}] \cdot \mathbb{E}[f(X) \mid \mathcal{F}]. \quad (11)$$

We note that

$$\mathbb{E}[f(X) \mid \mathcal{F}] = N_0(m) - 2\mathbb{E}[T_0(m) \mid \mathcal{F}] = N_0(m) - N_0(m) = 0$$

from (9). Additionally, $ED_1(m) \geq 0$ is $\mathcal{F}$ -measurable and $h(x) \geq 0$. Using the inequality in (11) gives us

$$\begin{aligned} \mathbb{E}[g(X) \mid \mathcal{F}] &= \mathbb{E}[h(X)f(X) \mid \mathcal{F}] + ED_1(m) \cdot \mathbb{E}[h(X) \mid \mathcal{F}] \\ &\geq \mathbb{E}[h(X) \mid \mathcal{F}] \cdot \mathbb{E}[f(X) \mid \mathcal{F}] + ED_1(m) \cdot \mathbb{E}[h(X) \mid \mathcal{F}] \\ &= 0 + ED_1(m) \cdot \mathbb{E}[h(X) \mid \mathcal{F}] \geq 0. \end{aligned}$$

Hence, by (10), $\mathbb{E}[\text{FDR}_B(m) - \text{FDP}(m) \mid \mathcal{F}] \geq 0$. As γ^* is $\mathcal{F}$ -measurable, plugging in $m = \gamma^*$ gives

$$\mathbb{E}[\text{FDR}_B(\gamma^*) - \text{FDP}(\gamma^*) \mid \mathcal{F}] \geq 0$$

and using the tower property,

$$\mathbb{E}[\text{FDR}_B(\gamma^*) - \text{FDP}(\gamma^*)] = \mathbb{E}[\mathbb{E}[\text{FDR}_B(\gamma^*) - \text{FDP}(\gamma^*) \mid \mathcal{F}]] \geq \mathbb{E}[0] = 0.$$

Therefore, $\mathbb{E}[\text{FDP}(\gamma^*)] \leq \mathbb{E}[\text{FDR}_B(\gamma^*)]$. $\square$

3.2 Bounding Under the p -value Setting

We next make the following assumption on the p -values of the true null hypotheses.

Assumption 2. If $i \in \mathcal{H}_0$, then $p_i \sim \text{U}(0, 1)$ independently of $\{p_j : j \neq i\}$.

Then, Theorem 1 still holds if we transform the p -values via Algorithm 2 and if we define $\mathcal{F}$ as in (6). We formally state this as a corollary.

Corollary 1. Assume that the p -values satisfy Assumption 2, and apply the transformation in Algorithm 2. If $\gamma^* := \max\{m : \omega(m) \leq \alpha\}$, then

$$\mathbb{E}[\text{FDP}(\gamma^*)] \leq \mathbb{E}[\text{FDR}_B(\gamma^*)].$$

Proof. All that is required for this proof is to show that Assumption 2 implies Assumption 1 under the transformation in Algorithm 2.

Take any $i \in \mathcal{H}_0$ with $L_i^o = 1$. Then, $p_i \sim \text{U}(0, 1)$ and from how the pre-flipped scores $\widetilde{W}_{1i}$ and $\widetilde{W}_{2i}$ are defined, we have $\widetilde{W}_{1i} = \widetilde{W}_{2i} = \min\{p_i, 1 - p_i\}$.

Denote this value by $w \in [0, \tau)$, so that $p_i \in \{w, 1 - w\}$. By uniformity, it follows that

$$\mathbb{P}(p_i = w \mid \widetilde{W}_{2i} = w, L_i^o = 1) = \mathbb{P}(p_i = 1 - w \mid \widetilde{W}_{2i} = w, L_i^o = 1) = \frac{1}{2} \quad \text{a.s.} \quad (12)$$

Note that as $p_i \sim \text{U}(0, 1)$ is uniform, then $\mathbb{P}(p_i = \tau) = 0$. Also, only one of $\{w, 1 - w\}$ can be in $[0, \tau)$. Without loss of generality, assume that $w < \tau$. Then, $p_i = w \iff L_i^h = 1$ and $p_i = 1 - w \iff L_i^h = -1$.

Therefore, conditional on the score W_2 (as there is a bijection from $\widetilde{W}_{2i}$ to W_{2i}) we have

$$\mathbb{P}(L_i^h = 1 \mid W_{2i}, L_i^o = 1) = \mathbb{P}(L_i^h = -1 \mid W_{2i}, L_i^o = 1) = \frac{1}{2} \quad \text{a.s.}$$

Rewriting the σ -field $\mathcal{F}$ as

$$\mathcal{F} = \sigma\left(W_{1i}, W_{2i}, L_i^o, \underbrace{\{W_{1j}, W_{2j}, L_j^o\}_{j \neq i}, \{j\}_{j \in \mathcal{H}_1}, \{L_j^h\}_{j \in \mathcal{H}_1}}_{=: \mathcal{G}}\right),$$

we note that by Assumption 2, p_i is independent of $\mathcal{G}$ and hence $(W_{1i}, W_{2i}, L_i^h, L_i^o)$ is independent of $\mathcal{G}$. Then by the independence of L_i^h and $\mathcal{G}$ and that $W_{1i} = W_{2i}$,

$$\mathbb{P}(L_i^h = 1 \mid \mathcal{F}) = \mathbb{P}(L_i^h = 1 \mid \mathcal{G}, W_{1i}, W_{2i}, L_i^o = 1) = \mathbb{P}(L_i^h = 1 \mid W_{2i}, L_i^o = 1) = \frac{1}{2} \quad \text{a.s.}$$

An analogous argument also shows that

$$\mathbb{P}(L_i^h = -1 \mid \mathcal{F}) = \mathbb{P}(L_i^h = -1 \mid W_{2i}, L_i^o = 1) = \frac{1}{2} \quad \text{a.s.}$$

It follows that

$$\mathbb{P}(L_i^h = 1 \mid \mathcal{F}) = \mathbb{P}(L_i^h = -1 \mid \mathcal{F}) \quad \text{a.s.}$$

for any $i \in \mathcal{H}_0$ with $L_i^o = 1$. Therefore, Assumption 1 holds, and under this $\mathcal{F}$ the result follows by Theorem 1. $\square$

We note that the equality in (12) can be weakened to one-sided $\leq$ bounds under the weaker assumption of **super-uniformity** of true null p -values. That is, the distribution of the true null p -values stochastically dominates the $U(0, 1)$ distribution.

Additionally, the independence assumption can be weakened to just requiring the joint distribution of the true nulls, conditioned on the false nulls, to be invariant under the transformation $p \mapsto 1 - p$ applied to any subset of the true nulls. This condition is satisfied by independent p -values and several dependent structures.

4 Simulation Studies

In this section, we assess the performance of the FDR Bound procedure in comparison to the classical FDR control procedures: Benjamini-Hochberg (BH), Storey’s q -value procedure, and Benjamini-Yekutieli (BY).

We compute the FDP and power using the true/false null labels for each p -value we generate over our simulations. To do this, we find the discoveries for each procedure, define the false discoveries (FD) as the rejected true nulls, and set

$$\text{FDP} = \frac{\#\text{FD}}{\max\{\#T_S, 1\}}.$$

Here, T_S denotes the set of discoveries for the procedure under consideration. We define the power as the proportion of false nulls that we reject:

$$\text{Power} = \frac{\#T_S - \#\text{FD}}{\#\mathcal{H}_1}.$$

We generate M sets of n p -values, then compute the FDP and power for each set, and take averages over the M iterations to obtain the empirical FDR and power. We do this for the thresholds $\alpha \in \{0.01, 0.02, \dots, 0.25\}$.

4.1 Independent p -values

In our first simulation, the true nulls are independent and identically distributed (i.i.d) $U(0, 1)$ observations, and the false nulls are i.i.d $\text{Beta}(0.4, 4)$ observations. In other words, we generate the p -values from the mixture distribution

$$f(x) = \pi_0 U(0, 1) + \pi_1 \text{Beta}(0.4, 4).$$

We perform $M = 100$ iterations to estimate the empirical FDR and power using the FDR Bound procedure, varying the sample size n , τ (as defined in [Section 2.2](#)), and c (as defined in the introduction of [Section 2](#)).

For each simulation, we run through the range of values of one parameter and keep the other two parameters as their default values, as summarised in Table 3. In addition, we keep the mixture weight $\pi_0 = 0.8$ throughout these simulations.

Table 3: Simulation parameters for independent p -values

Parameter	Default	Simulation Values
n	1000	1000, 5000, 10000
τ	0.25	0.1, 0.175, 0.25
c	1	0, 0.5, 1, 2

In Figure 1, we can see some subtle patterns in how these parameter changes affect our FDR and power. In particular,

- An increase in n from 1000 to 5000 leads to a higher overall FDR, which makes sense intuitively, as there is a greater number of both true and false null hypotheses. The change from 5000 to 10000 is less noticeable. The power seems to be more or less the same over each value of n .
- The results across each value of τ are very similar, so τ has little effect on the FDR. This makes sense, as while we generate fewer Targets and Estimating Decoys, the r we include in the selection function ω changes based on τ as well, offsetting this effect.
- There is a more noticeable difference across the c simulations for smaller α values. In particular, a larger c seems to lead to a smaller FDR. This is because as c increases, so does $\omega(m)$, and hence a lower cutoff τ^* . This produces fewer discoveries and thus lower FDR and power.

We also compare our procedure’s FDR and power with those of BH, Storey, and BY. We provide Table 4 as a summary of the results, including the empirical FDR, the average power, and the average number of discoveries for the four procedures. Note that we performed these simulations with $M = 500$ iterations.

Table 4: Simulation results for independent p -values at FDR threshold $\alpha = 0.10$

Procedure	FDR	Power	Discoveries
FDR Bound	0.091	0.251	56.228
BH	0.077	0.227	49.568
BY	0.010	0.058	11.788
Storey	0.100	0.269	60.314

Figure 2 shows the FDR and power of each procedure. Our procedure’s FDR hovers around the diagonal line, performing similarly to Storey, while BH and BY are more

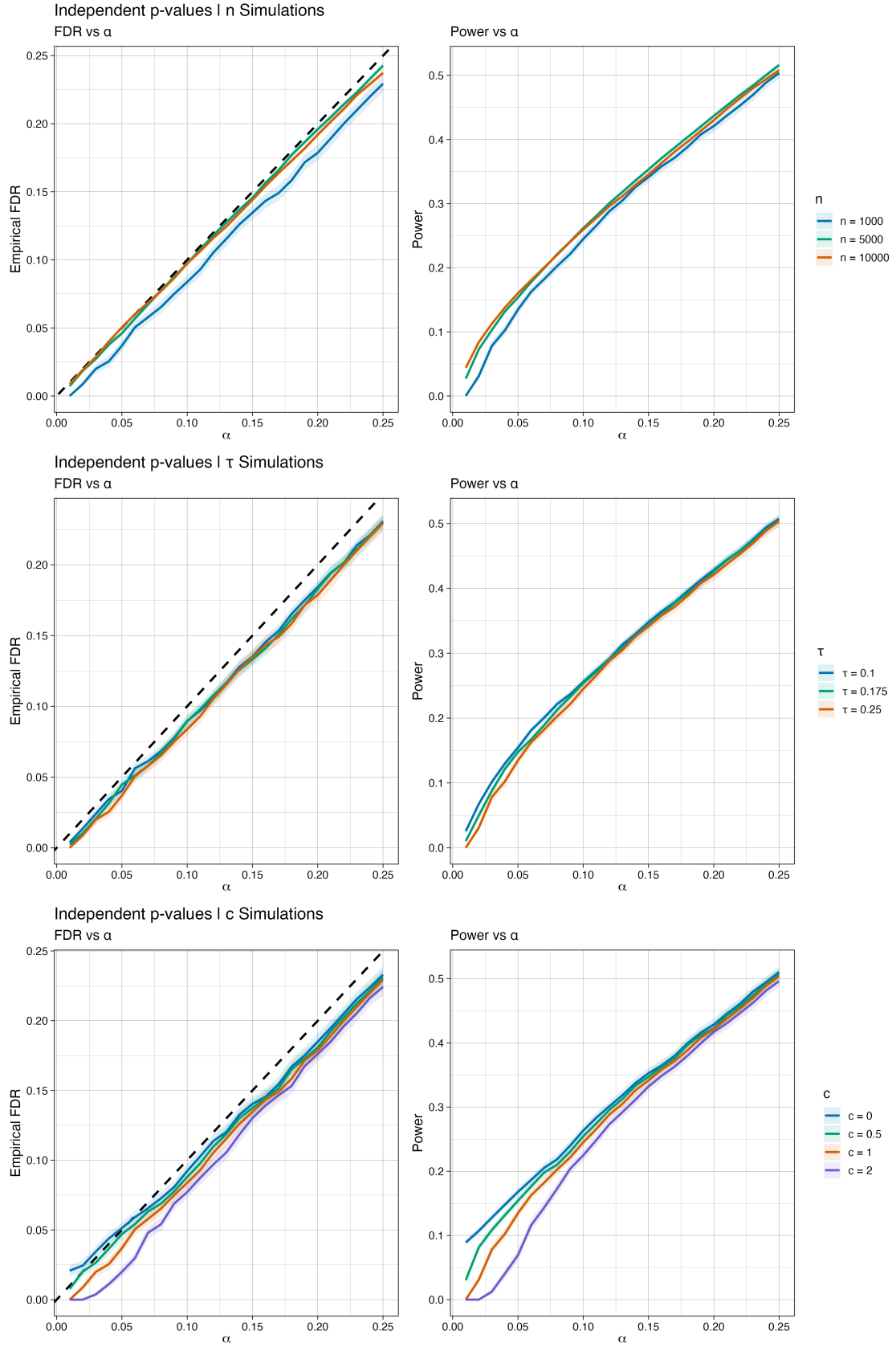


Figure 1: Results of n , τ , and c simulations

conservative. Our power is similar to that of Storey and BH, and we also control the FDR on average in these simulations.

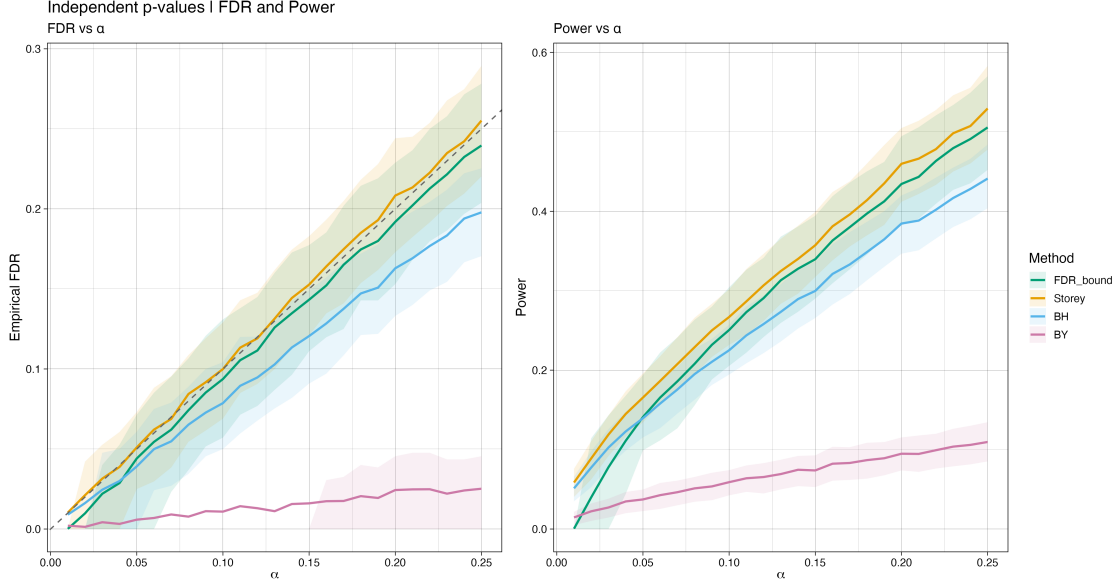


Figure 2: Simulation results for FDR and power for independent p -values; shaded regions are 25th/75th percentile bands

4.2 Dependent Structures

We consider various cases of local dependence among true nulls, independent of false nulls, while still being marginally $U(0, 1)$. We are particularly interested in cases where Storey or BH fails, to compare how our procedure performs in these scenarios.

For our first example of local dependence, we partition the n hypotheses into B blocks, then for each block generate $U_b \sim U(0, 1)$ and set every true null in block b to have U_b as its p -value; false nulls are i.i.d $\text{Beta}(0.4, 4)$. We denote the block size by k , so that $Bk = n$.

This local dependence between true nulls leaves them marginally $U(0, 1)$; however, the indicator functions $\mathbf{1}_{\{p_i > \lambda\}}$ have perfect correlation in each block, implying upper tail dependence, which can cause Storey to violate FDR control. Table 5 summarises the simulation setup for this structure, which we call the **Null Clumps**. For the FDR Bound parameters τ and c , we keep them as before, with $\tau = 0.25$ and $c = 1$.

Table 6 provides a summary of the results under this set of parameters at the FDR threshold $\alpha = 0.10$. However, at this level, it is not clear that there is a major violation of FDR control. Our performance is also quite similar to that of Storey's.

It is clearer in Figure 3 that for some larger values of α , our procedure and Storey still behave very similarly, and also fail to control the FDR for some values of α ,

Table 5: Simulation parameters for Null Clumps

Parameter	Description	Value
n	Sample Size	1000
M	Iterations	1000
π_0	True Null Mixture Weight	0.8
k	Block Size	50

 Table 6: Simulation results for Null Clumps at FDR threshold $\alpha = 0.10$

Procedure	FDR	Power	Discoveries
FDR Bound	0.113	0.838	192.560
BH	0.076	0.813	177.484
BY	0.008	0.497	100.504
Storey	0.117	0.853	196.713

albeit only mildly. BH has a lower FDR and still controls it. In terms of power, our procedure performs similarly to BH and Storey; BY remains the least powerful.

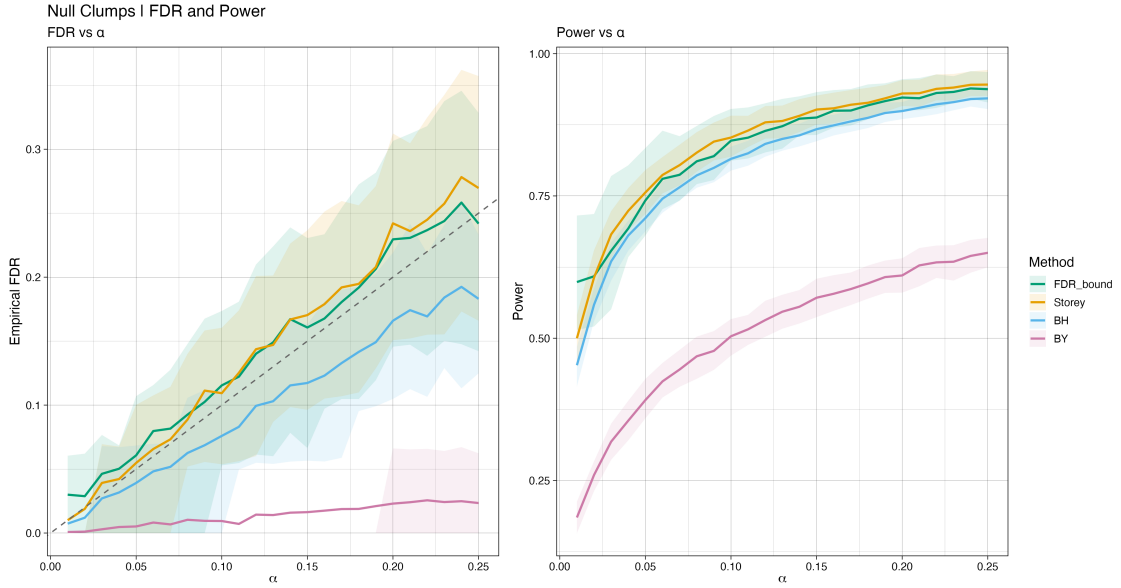


Figure 3: Simulation results for FDR and power for Null Clumps

We include one more example that demonstrates a case where both BH and Storey fail. This structure is adapted from Guo and Rao [7] and is constructed as a “worst-case” for BH.

We made a few small tweaks, such as clipping the α parameter to its maximum feasible value

$$\alpha_{\max} = \left\{ \frac{m - m_0}{m} + \frac{m_0}{m} \sum_{j=1}^{m_0} \frac{1}{j} \right\}^{-1}$$

with m_0 and m defined in Table 1, and replacing false nulls by i.i.d Beta(0.4, 4) observations, allowing them to be independent of true nulls. As a result of our first change, the distribution of true nulls is not always $U(0, 1)$; however, it remains super-uniform.

This structure is quite artificial and uses the FDR threshold α as a parameter. As a result, it should be treated as a very adversarial case of local dependence designed for failure in FDR control. There are no additional parameters besides (n, M, π_0) , which we summarise in Table 7. The FDR Bound parameters $\tau = 0.25$ and $c = 1$ are kept as before.

Table 7: Simulation parameters for Guo-Rao’s Worst Case

Parameter	Description	Value
n	Sample Size	1000
M	Iterations	1000
π_0	True Null Mixture Weight	0.8

We start by simulating at the FDR threshold $\alpha = 0.10$ as before, and in Table 8 we see that our procedure, BH, and Storey all fail to control the FDR, with FDR exceeding 0.2. BY still controls the FDR very well, as this structure was designed with BY in mind.

Table 8: Simulation results for Guo-Rao’s Worst Case at FDR threshold $\alpha = 0.10$

Procedure	FDR	Power	Discoveries
FDR Bound	0.267	0.782	346.692
BH	0.232	0.304	153.732
BY	0.011	0.062	12.580
Storey	0.228	0.296	157.444

Figure 4 confirms this, as the FDRs of our procedure, BH and Storey are well beyond the diagonal. Additionally, it seems that our procedure performs the worst for small values of α , although the FDR seems to peak, likely at the feasibility point of α , as beyond that the true nulls are strictly super-uniform.

Our procedure attains much higher power here and appears to converge more quickly with increasing α than BH and Storey, which have similar power.

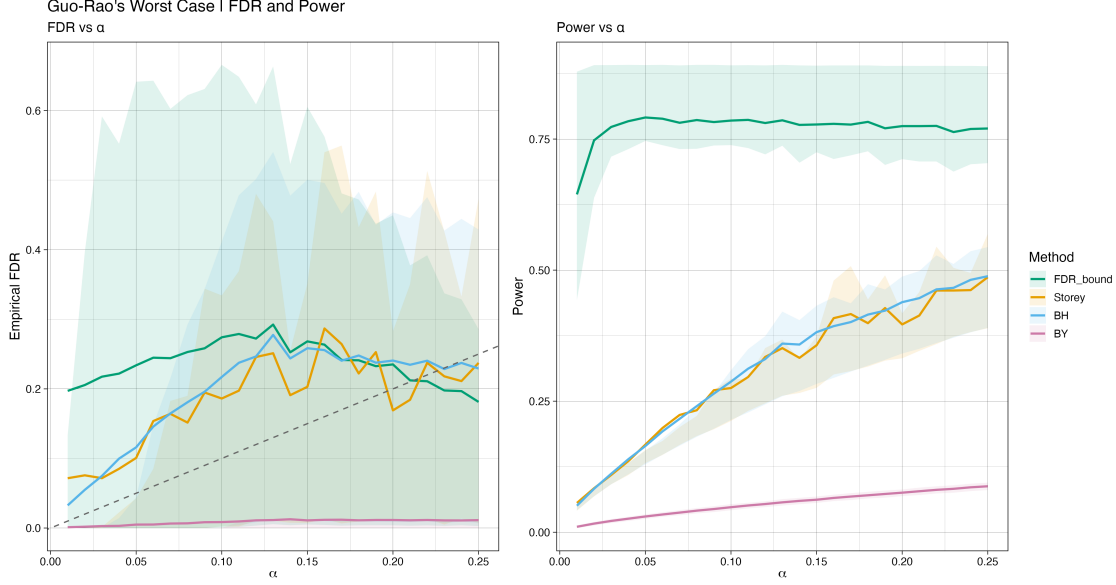


Figure 4: Simulation results for FDR and power for Guo-Rao's Worst Case

4.3 Comparing the FDR Bound with Empirical FDR

Next, we want to empirically determine whether the results of [Section 3.1](#) hold by checking the difference between the average FDR Bound and the average FDP over the M simulations.

We did this for all the simulations we performed and found that, for *most* simulations, our FDR Bound exceeds the FDP on average. However, there are cases where this does not hold, particularly where the true nulls are not independent of the false nulls. One such example is when *every* p -value depends on the same latent variable $F \sim \mathcal{N}(0, \gamma^2)$ for some $\gamma \in (0, 1)$. For each $i \in [n]$, we generate the Z -score

$$Z_i = F + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1 - \gamma^2).$$

For false nulls, we add a factor of μ (for example, $\mu = 2$) to Z_i . Then, the p -values are given as $p_i = 1 - \Phi(Z_i)$ where Φ is the CDF of the $\mathcal{N}(0, 1)$ distribution. Then, both true and false nulls are dependent on F . We denote this structure as the **Common Dominant Factor**.

Figure 5 shows the Average FDR Bound minus Average FDP plots of the four structures that we discussed in this section. For the first three structures, we note that we do not always exceed the FDP for all α , but this is not a major concern as it is not systematic. For the Common Dominant Factor, we see that, in Figure 5, for the first few values of α , the FDR Bound is systematically under the FDP.

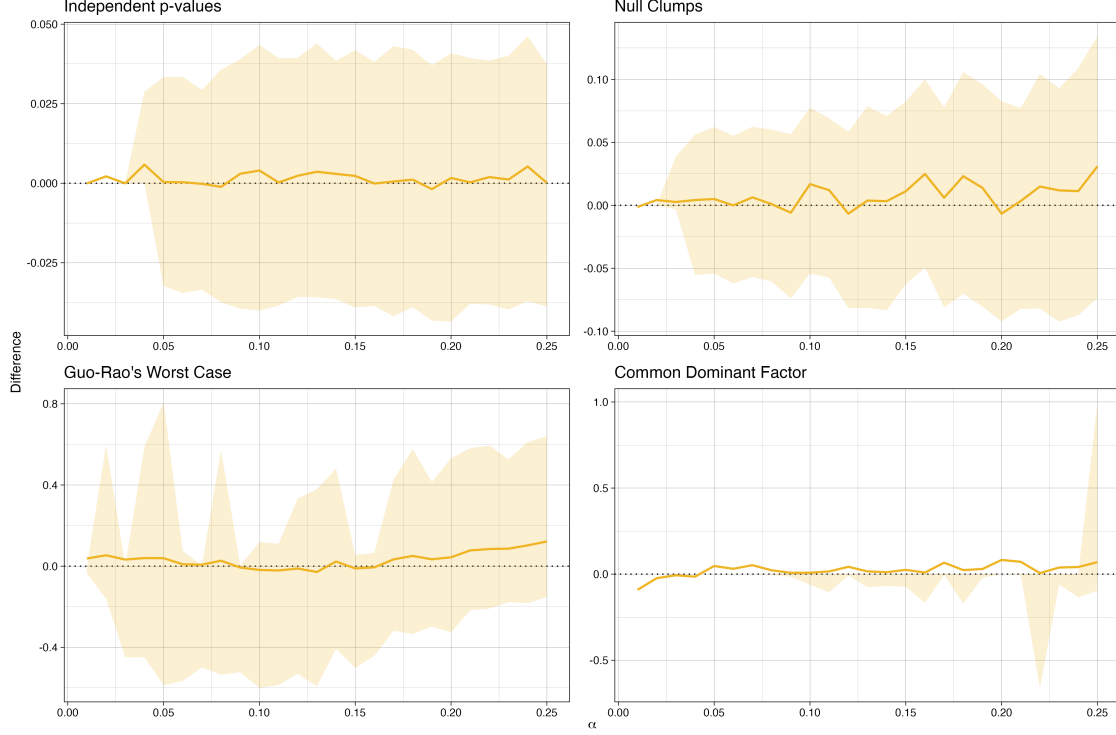


Figure 5: Average FDR Bound minus Average FDP plots for the four structures in Section 4, with 25th and 75th percentile bands included

4.4 Clipping the FDR Bound

We typically clip the FDR Bound at 1 in these simulations; that is, we report and use $\min\{\text{FDR}_B, 1\}$. For example, Figure 5 compares the *clipped* FDR Bound against the FDR. We now analyse whether clipping changes the reported value of the FDR Bound.

Figure 6 shows that under the independent p -values and Null Clumps cases, the clipped and raw FDR Bounds coincide. However, Figure 7 shows that in the Guo-Rao and Common Dominant Factor cases, we obtain a much larger raw FDR Bound (far exceeding 1) and hence the difference between the FDR Bound and FDR is largely positive.

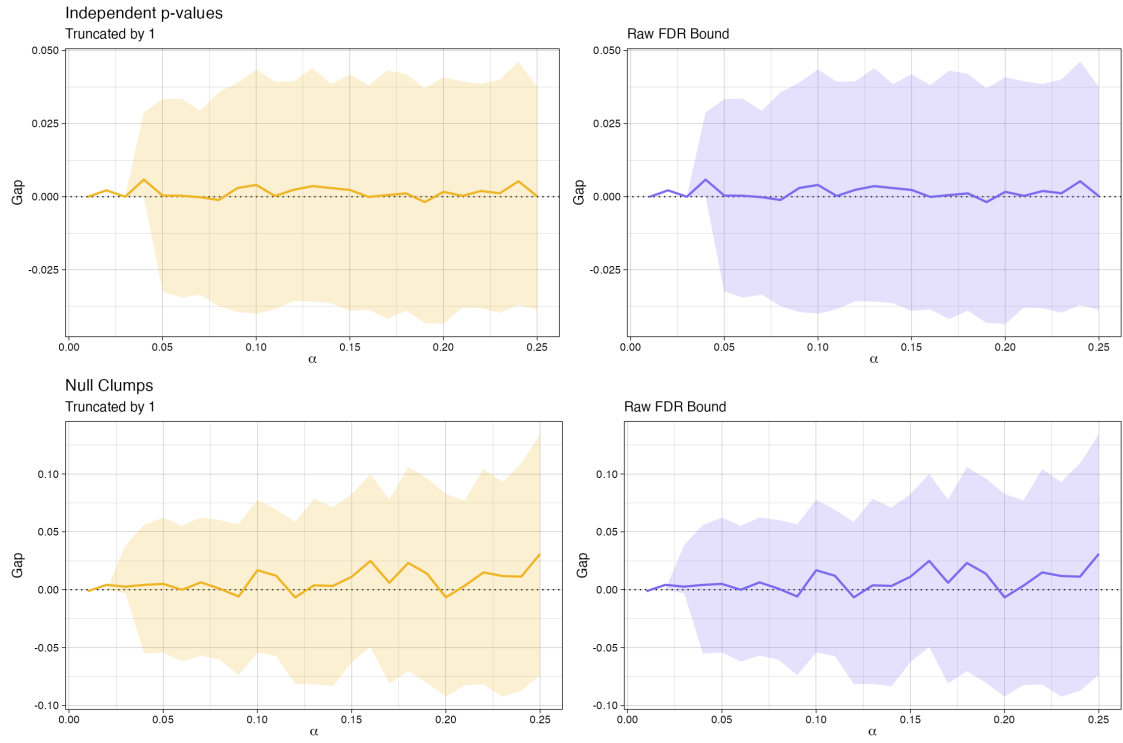


Figure 6: Effects of clipping the FDR Bound at 1 on independent p -values and Null Clumps

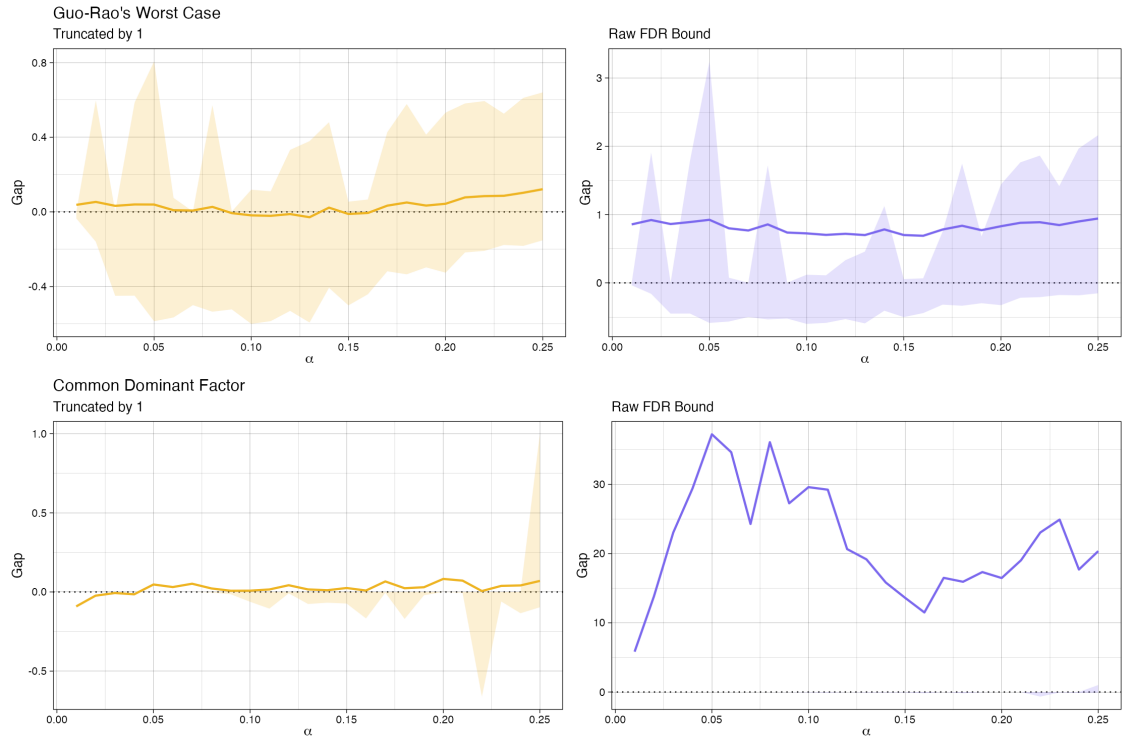


Figure 7: Effects of clipping the FDR Bound at 1 on Guo-Rao's Worst Case and Common Dominant Factor

5 Conclusion

This report has developed a procedure that provides a rigorous upper bound on the FDR under certain assumptions. We discussed the FDR Bound procedure and proved its bounding property in both a TDC setting and an independent p -value setting, though we remark that our theoretical results hold under weaker assumptions.

We conducted studies on specific parameters of the FDR Bound and found only marginal differences in performance. We also conducted simulation studies under different p -value structures to compare our procedure's FDR and power against those of classical FDR control procedures, focusing on specific cases where these procedures fail.

We found cases where our procedure fails to **control** the FDR under local dependence of true nulls (while remaining independent of false nulls), and a case where it fails to **bound** the FDR under arbitrary dependence between true and false nulls.

The failure to bound under arbitrary dependence can be considered a potential limitation of our procedure. Another limitation is that our implementation of Storey's q -value procedure tends to return invalid (NA) values in simulations. We based our conclusions about Storey's procedure only on the valid iterations, which may have skewed our interpretations. Additionally, many of our simulation studies, such as the Guo-Rao simulation, use very contrived dependent structures.

As an avenue for future work, we may begin by applying our procedure to real data to assess its performance in less pathological settings. Moreover, we can investigate the theory behind asymptotic FDR control and compare our procedure with the empirical Bayes/local FDR framework introduced in Efron [4] and with other procedures in the literature.

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